1 A thin sheet of cardboard is supported by four springs. Figure 1.1 shows a side view of the arrangement.

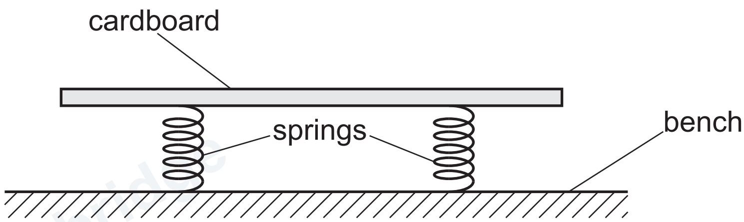

Figure 1.1

A steel ball of radius r is dropped onto the cardboard. The speed of the ball as it hits the cardboard is v.

As the ball hits the cardboard, the springs compress. The maximum compression of the springs is c.

Several steel balls of different radii are available.

It is suggested that c is related to r by the relationship

$$\frac{v^2}{c^2} = \frac{\alpha}{\pi \rho r^3} + Z$$

where ρ is the density of steel, and α and Z are constants.

Plan a laboratory experiment to test the relationship between c and r.

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine values for α and Z.

In your plan, you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

Diagram

[15]